

Lesson Plan — Lavanya Thondavada

Date: 2026-09-10 **Session length:** 120 minutes **Skill(s):** Nonlinear equations in one variable and systems of equations in two variables

Objectives

- Solve a single nonlinear equation (quadratic, absolute value, or radical) using the method that fits its structure, without wasting time forcing a factoring attempt that isn't there.
- Recognize when a question only asks "how many solutions" and answer it without fully solving (discriminant, or counting graph intersections).

Warm-up (10 min)

Two official questions from a previous skill (Nonlinear functions), for retrieval practice.

1. [Nonlinear functions] beca03de — A rectangle has a length that is 15 times its width. The function $y = (15w)(w)$ represents this situation... Which of the following is the best interpretation of $15w$ in this context?
2. [Nonlinear functions] 71b74a44 — $y = 7,400(0.87)^x$ estimates the value, in dollars, of a piece of equipment x years after purchase. What was the estimated value at the time it was purchased?

Direct instruction (30 min)

Link the notes page: `students/lavanya-thondavada/notes/nonlinear-equations-in-one-variable-and-systems-of-equations-in-two-variables.md`

- Walk through "The method" section: is it one equation or a system? Does it ask for the solutions or just how many there are?
- Work the discriminant example (332cd67b) and the "system from a graph" example (a5663025) live, side by side, to show the algebra shortcut vs. the graph-counting shortcut answer the same question type two different ways.

Desmos move demo

Take 7f81d0c3 ($x^2 - x - 1 = 0$). Type `y1=x^2-x-1` and `y2=0`, click both x-intercepts — Desmos gives $(1 \pm \sqrt{5})/2$ as decimals instantly, which you can then match back to the exact-form answer choices.

Contrast with a system question (d0a7871e): graph `y1=x+1` and `y2=x^2+x`, click the intersection(s) directly instead of solving algebraically.

Guided practice (65 min)

6 official questions, easy to hard.

1. (Easy) 3c95093c — $6x - 9y > 12$. Which inequality is equivalent?
2. (Easy) 1e003284 — $x = 49$, $y = \sqrt{x} + 9$. Value of y at the intersection?
3. (Medium) d0a7871e — $y = x+1$, $y = x^2+x$. Which x could be a solution?
4. (Medium) 4e18fc5d — $v = -w/(150x)$. Express w in terms of v and x .
5. (Hard) 3a9d60b2 — $2|4-x| + 3|4-x| = 25$. Positive solution?
6. (Hard) fc3d783a — Line $2y=4.5$ meets parabola $y=-4x^2+bx$ exactly once, $b>0$. Find b .

Exit ticket (15 min)

Two questions, done independently.

1. b4acba95 — $x^2 - 12x + 27 = 0$. How many distinct real solutions?
2. ba0edc30 — $x^2 - 2x - 9 = 0$. One solution is $1+\sqrt{k}$. Find k .

Homework assigned

students/lavanya-thondavada/homework/2026-09-10-nonlinear-equations-hw.md — due 2026-09-17 (next session).

Notes for me after the session

- What clicked / what didn't:
 - Adjust next lesson:
 - Update mastery table? (Y/N)
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Timing check: $10 + 30 + 65 + 15 = 120$ minutes.